Tenstorrent Galaxy fleet deployment map

Source type: PDF · 4 pages · data-room extract (scrubbed; the joint-venture counterparty is referred to only as 'VVIP Sovereign JV').

D E P L O Y M E N T M A P · V 1 P L A N N I N G D R A F T · 2 3 A U G 2 0 2 6 · I N T E R N A L
23-Galaxy Tenstorrent Inference Fleet
Containerised serving of Kimi K3 (flagship) and DeepSeek V4-Flash-Vision (workhorse) at a blended 4·5M
tokens per minute · hardware, container stack, capacity math and rollout
BLACKHOLE CHIPS
23 TB
GDDR6 @ 368 TB/S AGG.
529 PF
BLOCK FP8 COMPUTE
~200·230 kW
AVG IT LOAD · 6 RACKS
$2.9·4.4M
ALL-IN CAPEX EST.

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EXECUTIVE SUMMARY
FLEET AT A GLANCE
POOL
UNITS MODEL (BUILD)
PARAMS TOT /
ACTIVE
PRECISION
WEIGHTS REPLICAS
ROLE
Flagship
Kimi K3 (27/07 weights)
2.8T / ~50B
MXFP4
QAT
~1.4 TB
4× 2-unit
pairs
Deep reasoning & agents, 1M ctx, native
vision
Workhorse 13
DeepSeek V4-Flash-0731 ·
Vision-Exp
284B / 13B
FP4 + FP8
~167 GB
13× single
High-volume agentic, doc + chart vision,
1M ctx
Reserve
Floating (either)
Spare +
canary

Failover, canary releases, aux
(embeddings, OCR, guardrails)
Target envelope.

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Target envelope. 4·5M TPM blended (input + output, fleet-wide) = 67·83k tok/s sustained.
Assumed traffic mix 80·85% input / 15·
20% output, typical of agentic + document-vision loads; that implies ~500·800 concurrent generations at 25·35 tok/s/user with prefill
bursts well above 0.5M tok/s. Own-cost at target utilisation models to ~$0.7·0.9 per M blended tokens over 3 years.

at target utilisation models to ~$0.7·0.9 per M blended tokens over 3 years. If ·4/5M TPM· meant per-model rather than blended total, the fleet roughly doubles · flag before ordering
Buy 23× Galaxy Blackhole (6U air-cooled, 32 chips, 1 TB GDDR6 @ 16 TB/s, 23 PFLOPS FP8, 8·10 kW, GA since Apr 2026) ·
hardware $2.5·3.7M at $110k GA-reported vs $160k current list; negotiate supercluster rates through the Tenstorrent relationship
Placement is forced by memory: Kimi K3·s ~1.4 TB MXFP4 weights exceed a single 1 TB system · 4 replicas on 2-unit tensor-

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parallel pairs (8 units); V4-Flash·s ~167 GB fits one unit with ~830 GB left for KV · 13 single-unit replicas; 2 units stay as hot
spare + canary
Container stack is off the shelf: tt-kmd driver · TT-Metalium runtime · vLLM Tenstorrent plugin, packaged by tt-inference-
server as Docker images with a Helm chart on Kubernetes, fronted by an OpenAI-compatible gateway with per-tenant token
metering
Capacity is decode-gated: conservative kernels model 2.6·4.0M TPM blended at an 80/20 input/output mix · speculative

kernels model 2.6·4.0M TPM blended at an 80/20 input/output mix · speculative
decoding (V4-Flash ships its DSpark module; K3 has MTP) is the lever that clears the target at a modeled 4.3·6.8M TPM
Two gates before fleet-wide commit: neither model is on Tenstorrent·s validated list yet (Kimi K2 and DeepSeek V3/R1 are) ·
run a 1-unit port benchmark then a 4-unit pilot before the full PO; and V4-Flash-Vision-Exp is API-only since 21/08 with no first-
party weights on HF · launch on V4-Flash-0731 (MIT) and swap when vision weights land

Hardware · buy list, power and racks
02 · HARDWARE & FACILITY

weights land

Hardware · buy list, power and racks
02 · HARDWARE & FACILITY
GALAXY BLACKHOLE · UNIT SPEC (OFFICIAL)
Accelerators
32× Blackhole ASIC (RISC-V Tensix cores)
Compute
23 PFLOPS Block FP8 per system
On-chip SRAM
6.2 GB @ 2.9 PB/s

DRAM
1 TB GDDR6 @ 16 TB/s aggregate
Internal fabric
10× 400 GbE per ASIC · 32 TB/s Ethernet
mesh, no proprietary interconnect
Scale-out
Up to 56× 800 GbE QSFP-DD · 11.2 TB/s
(populate only what the topology needs)
Host
1× AMD EPYC 9004 (·32c), ·576 GB DDR5
ECC, 2× 960 GB NVMe (+ optional expansion)
Form factor
6U rackmount, air-cooled, front-to-back
Power

optional expansion)
Form factor
6U rackmount, air-cooled, front-to-back
Power
8·10 kW avg / 12 kW max
Price
$160k list (Aug 2026); $110k reported at Apr GA;
4-unit supercluster $640k list / $440k at GA
DOES IT FIT IN MEMORY? (THIS DRIVES THE
PLACEMENT)
MODEL
WEIGHTS FIT
KV HEADROOM
Kimi K3
2.8T MXFP4
~1.4 TB
> 1 TB · 2-
unit pair (2
TB)
~0.6 TB/pair; KDA linear attention
keeps KV small · hundreds of
100k+ streams
V4-Flash
284B FP4/
FP8
~167 GB
Single unit
~830 GB; CSA/HCA-compressed
KV · thousands of doc-length
streams, 1M-ctx jobs OK

GB; CSA/HCA-compressed
KV · thousands of doc-length
streams, 1M-ctx jobs OK
Host NVMe (1.92 TB stock) holds one K3 shard (~700 GB) + V4 weights +
OS; order the internal expansion on K3 hosts for multi-model caching and fast
re-shard
FLEET BILL OF MATERIALS (ESTIMATES)
ITEM
QTY EST. COST
Galaxy Blackhole systems
$2.53·

3.68M
800G leaf switches (32-port) for pair links + spine
$90·140k
400G ToR / mgmt switches
$30·50k
Optics + DACs (800G pair DACs, 400G uplinks)
lot
$60·120k
Racks, 3-phase PDUs, containment (40·50 kW/rack)
$50·90k
Weights/artifact storage node (~20 TB NVMe, MinIO/NFS

(40·50 kW/rack)
$50·90k
Weights/artifact storage node (~20 TB NVMe, MinIO/NFS
+ registry)
$40·80k
Spares kit (PSUs, fans, optics, 1 host board)
$50·100k
Integration, cabling, burn-in
$80·150k
Total capex
$2.9·4.4M
FACILITY ENVELOPE
IT load
184·230 kW avg, 276 kW peak (units only)
Racks
6 · five with 4× 6U + ToR, one with 3× +
storage/gateway
Cooling
Air, front-to-back; needs a 40·50 kW/rack high-
density row or rear-door heat exchangers
Facility power
240·345 kW at PUE 1.3·1.5 · 2.1·3.0 GWh/yr
Energy cost
$190·360k/yr at $0.09·0.12/kWh (UAE
commercial band)
Colo alternative

cost
$190·360k/yr at $0.09·0.12/kWh (UAE
commercial band)
Colo alternative
$150·250 per kW-IT/month · $415·690k/yr all-
in
Bandwidth
2× diverse 10·100G internet/peering; all
inference traffic stays in-hall
Sequencing note. Order at P0 start: system lead + integration
realistically 6·10 weeks post-GA · the fleet-build phase is supply-
gated, not engineering-gated.

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gated, not engineering-gated. 800G optics are the second-longest lead
item; pairs can run on DACs day 1
RACK & POOL MAP · 23 UNITS + STORAGE ACROSS 6 RACKS, POOLS COLOUR-CODED
R1 · K3 · ·40 kW

ToR / mgmt
K3 pair A · unit 1
K3 pair A · unit 2
K3 pair B · unit 1
K3 pair B · unit 2
R2 · K3 · ·40 kW
ToR / mgmt
K3 pair C · unit 1
K3 pair C · unit 2
K3 pair D · unit 1
K3 pair D · unit 2
R3 · V4 · ·40 kW
ToR / mgmt
V4-Flash r1
V4-Flash r2
V4-Flash r3
V4-Flash r4
R4 · V4 · ·40 kW
ToR / mgmt
V4-Flash r5
V4-Flash r6
V4-Flash r7
V4-Flash r8

r4
R4 · V4 · ·40 kW
ToR / mgmt
V4-Flash r5
V4-Flash r6
V4-Flash r7
V4-Flash r8
R5 · V4 · ·40 kW
ToR / mgmt
V4-Flash r9
V4-Flash r10
V4-Flash r11
V4-Flash r12
R6 · mixed · ·35 kW
800G leaf pair
V4-Flash r13
Reserve · canary
Reserve · hot spare
Storage + gateway
Kimi K3 pairs (8)
DeepSeek V4-Flash (13)
Reserve (2)
Direct 800G tensor-parallel pair link (DAC)
All units uplink 2×400G to leaf; API traffic north-south, TP traffic stays in-rack

Serving stack, then the capacity math
03 · SOFTWARE & CAPACITY
SOFTWARE STACK, BOTTOM TO TOP (ALL OPEN SOURCE)
LAYER
COMPONENT
DEPLOYMENT NOTES
Host & driver

BOTTOM TO TOP (ALL OPEN SOURCE)
LAYER
COMPONENT
DEPLOYMENT NOTES
Host & driver
Ubuntu 22.04/24.04, tt-kmd kernel module, tt-flash
firmware, hugepages, tt-smi, tt-topology
tt-installer bootstraps a node in one command; bake into a golden image
(Ansible/MAAS) for 23-node repeatability
Runtime
TT-Metalium / TT-NN kernel + op library; TT-Forge
(MLIR) compile path
Vendor line: ·90% of HuggingFace models just work· · true for functionality,
not speed; production graphs are hand-optimised per model
Serving
tt-inference-server: prebuilt Docker images, vLLM

per model
Serving
tt-inference-server: prebuilt Docker images, vLLM
Tenstorrent plugin (continuous batching, paged KV),
SGLang plugin, benchmark workflows
One serving container per unit exposing OpenAI-compatible endpoints; same
image runs LLM, VLM, embedding and TTS classes
Orchestration
Kubernetes (RKE2/kubeadm) + tt-inference-server
Helm chart
1 pod : 1 Galaxy, pinned by node label; K3 = 2-node tensor-parallel over
direct 800G links with pod affinity; devices exposed via privileged/hostPath
· no mature official K8s device plugin yet; close this gap with TT
Gateway

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Gateway
OpenAI-compatible router (LiteLLM / Envoy), per-
tenant quotas + token metering, priority classes,
retry & failover to reserve pool
Model-name routing: kimi-k3 · flagship pairs, v4-flash[-vision] ·
workhorse pool; slots straight into OnDemand ACR as a serving backend
Weights & data
MinIO/NFS weights store, per-host NVMe cache,
signed manifests
Blue-green weight rollouts: stage on reserve units · canary 5% · flip pool
by label; ~5·10 TB versioned store is plenty
Observability

5% · flip pool
by label; ~5·10 TB versioned store is plenty
Observability
Prometheus + Grafana; vLLM metrics (TTFT, TPOT,
queue depth), tt-smi exporter (power, temps, link
health), token accounting export
SLOs: p50 TTFT < 2 s at 32k ctx, p95 < 6 s; per-model TPM dashboards
feed the capacity re-plan
CAPACITY MODEL · DECODE IS THE CONSTRAINT, SO SIZE AGAINST IT
Method: decode is memory-bandwidth-bound · ceiling = BW ÷ active bytes/token; apply an
achieved-efficiency band; prefill is compute-bound and effectively

free at this mix.

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free at this mix. All figures are engineering estimates pending P0 benchmarks · no public
K3/V4-on-Galaxy numbers exist yet
POOL
REPLICASACTIVE BYTES /
TOKEN
BW PER REPLICA
CEILING
TOK/S
ACHIEVED (35·
55%)
POOL DECODE
TOK/S
V4-Flash (13B act., FP4/FP8 + KV
reads)
13× 1 unit~10 GB
16 TB/s
~1,600
560·880
7,300·11,400
Kimi K3 (50B act., MXFP4 + KV)
4× 2-unit
pair
~27 GB
32 TB/s (cross-node TP tax
· 25·40%)
~1,185
300·475
1,200·1,900
Fleet decode · base
8,500·13,300
·with speculative decode ×1.7 (V4
DSpark, K3 MTP)
14,500·22,600

8,500·13,300
·with speculative decode ×1.7 (V4
DSpark, K3 MTP)
14,500·22,600
Prefill check: V4-Flash ~26 GFLOP/token · tens to hundreds of thousands of tok/s per unit
burst; Tenstorrent demoed a 100k-token prompt in <4 s TTFT on a
671B-class model. Fleet prefill · 0.5M tok/s · not binding at an 80/20 mix
BLENDED TPM · SENSITIVITY ON THE TWO DRIVERS
OUTPUT SHARE OF MIX
BASE KERNELS
+ SPEC-DECODE (×1.7)
15% (prefill-heavy agents)
3.4·5.3M
5.8·9.0M
20% (planning case)
2.6·4.0M
4.3·6.8M ·
30% (chat-heavy)
1.7·2.7M
2.9·4.5M

Verdict.

(planning case)
2.6·4.0M
4.3·6.8M ·
30% (chat-heavy)
1.7·2.7M
2.9·4.5M
Verdict. The 23-unit fleet clears 4·5M blended TPM with speculative decoding on and an output share ·20%. If the mix runs hotter: move the 2 reserve units into the V4 pool (+~1,300 tok/s decode), shed K3 chat traffic to V4, or add one supercluster.

(+~1,300 tok/s decode), shed K3 chat traffic to V4, or add one supercluster. Quick math: 4.5M TPM = 75k tok/s; at 20% output that is 15k decode tok/s · inside the spec-decode band (14.5·22.6k)

Rollout, risks and the counter-case
04 · EXECUTION
ROLLOUT · GATE THE FLEET PO ON P0/P1 RESULTS
PHASE
WINDOW
SCOPE
EXIT CRITERIA
P0 Bring-up
Wk 0·2 (Sep)
1 unit: tt-installer, firmware, V4-Flash-0731 on vLLM-TT; benchmark vs the model on p3
·500 tok/s sustained decode; TTFT <5 s at 100k ctx; numbers within 30% of model
P1 Pilot
Wk 2·6 (Sep·Oct)

TTFT <5 s at
100k ctx; numbers within 30% of model
P1 Pilot
Wk 2·6 (Sep·Oct)
4-unit supercluster: K3 2-node TP port with TT engineering; load + chaos tests; K8s/Helm hardening
K3 pair ·300 tok/s; node-kill failover <60 s; 72 h soak clean
P2 Fleet build
Wk 6·10 (Oct·Nov)
All 23 racked + fabric, gateway, observability, weight pipeline, golden images
Burn-in complete; synthetic 3M TPM sustained 24 h
P3 Ramp
Wk 10·14 (Nov·Dec)
Production traffic to 4·5M TPM; spec-decode tuning; canary channel; Vision-Exp weight swap when published
SLOs green 2 consecutive weeks at target TPM

weight swap when published
SLOs green 2 consecutive weeks at target
TPM
TOP RISKS AND MITIGATIONS
RISK
L / I
MITIGATION
Neither K3 (KDA attention) nor V4-Flash (CSA/HCA
attention) is TT-validated; kernel/porting work of unknown
depth
High / High
P0/P1 gates before fleet PO; co-engineer with Tenstorrent; fallback that keeps
the fleet earning: Kimi K2 + DeepSeek V3/R1 are already TT-supported
V4-Flash-Vision-Exp weights not published (API-only
since 21/08)
Med / Med
Launch on 0731 text/tool build (MIT); community vision ports stay off-prod;
swap on first-party release

build (MIT); community vision ports stay off-prod;
swap on first-party release
Throughput model misses · no public K3/V4-on-Galaxy
benchmarks
Med / High
All numbers are ceilings × efficiency, not vendor quotes; P0 result re-sizes the
fleet (23 can flex to 16 or 28) before full commitment
List price drifted $110k · $160k in 4 months; supply
tightness
Med / Med
Lock supercluster pricing now through the TT relationship; PO at P0 start
Software maturity · independent reviews still flag rough
edges in the TT stack
Med / Med

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edges in the TT stack
Med / Med
Pin tested image versions; support contract; SGLang as the alternate serving
path
Single EPYC host per unit is a per-unit SPOF; 800G
optics lead times
Med / Low
K8s reschedules to reserve pool; spares kit on site; DACs for pair links day 1
COUNTER-ARGUMENTS · READ BEFORE
COMMITTING
ASSUMPTIONS I MADE
SOURCES
tenstorrent.com/hardware/galaxy · tenstorrent.com/en/solutions/llm-inference ·
theregister.com/2026/04/28/tenstorrent_galaxy_blackhole_ai_servers_ga ·

·
theregister.com/2026/04/28/tenstorrent_galaxy_blackhole_ai_servers_ga ·
hpcwire.com/off-the-wire/tenstorrent-announces-general-availability-of-galaxy-
blackhole-ai-system · wccftech.com/tenstorrent-vows-to-crush-everyone-
galaxy-blackhole-hits-350-tokens-on-deepseek-r1-undercut-nvidia-gb300-ai-
tco · github.com/tenstorrent/tt-inference-server · huggingface.co/deepseek-ai/
DeepSeek-V4-Flash (+ -0731, -Base) · api-docs.deepseek.com/news/

news260821 · openrouter.ai/deepseek/deepseek-v4-flash-vision-exp · huggingface.co/blog/ResterChed/kimi-k3-model-overview-mxfp4-quantization-

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huggingface.co/blog/ResterChed/kimi-k3-model-overview-mxfp4-quantization-
open-wei · huggingface.co/blog/ResterChed/deepseek-v4-flash-official-release · tomshardware.com (Kimi K3 weights release)

The API undercuts you on pure $/token: DeepSeek prices V4-Flash-Vision at $0.22/$0.66 per M tok · ~$0.31 blended vs ~$0.7·0.9 own-cost.

Vision at $0.22/$0.66 per M tok · ~$0.31 blended vs ~$0.7·0.9 own-cost.
The fleet case rests on sovereignty, air-gap and K3-class control · which is the actual thesis; concede the point for non-sovereign overflow traffic

A GPU cluster is the boring alternative: 4·6 B200/GB200-class nodes likely match this throughput on mature software day 1; TT wins claimed $/token (TT: $6 vs ~$30/M on 671B-class decode) and openness, loses ecosystem maturity · P0/P1 is the honest test

24 beats 23: six clean 4-unit superclusters price, rack and pair better; the

24 beats 23: six clean 4-unit superclusters price, rack and pair better; the
23rd unit buys nothing structural · consider 24 or drop to 22

Vision volume on the flagship: K3 already has native vision · if doc-vision dominates, some ·workhorse· traffic may belong on K3 pairs, changing the 8/13 split

·K3/3.1· = Kimi K3, 27/07 open weights; no K3.1 announced as of 23/08 · treated as a drop-in successor when it ships

·Vision 22/08· = DeepSeek-V4-Flash-Vision-Exp, API launch 21/08 UTC (22/08 GST)

4·5M TPM = blended input+output across the whole fleet, 80·85% / 15·20% mix

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Throughput figures are engineering estimates (BW ÷ bytes/token × 25·55%), pending P0 benchmarks; prices are public list/reported, pre-negotiation